



A “Monitor”

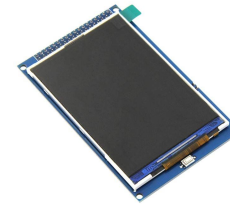
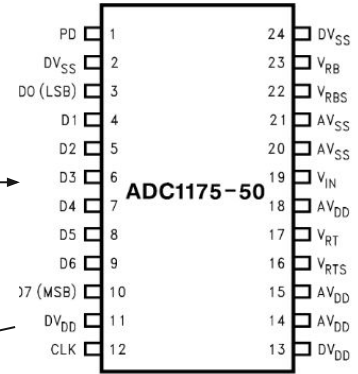
Trevor Sribar
4/24/2026



Project Overview

Take the HDMI input from a windows computer and display a grayscale image.

Hardware



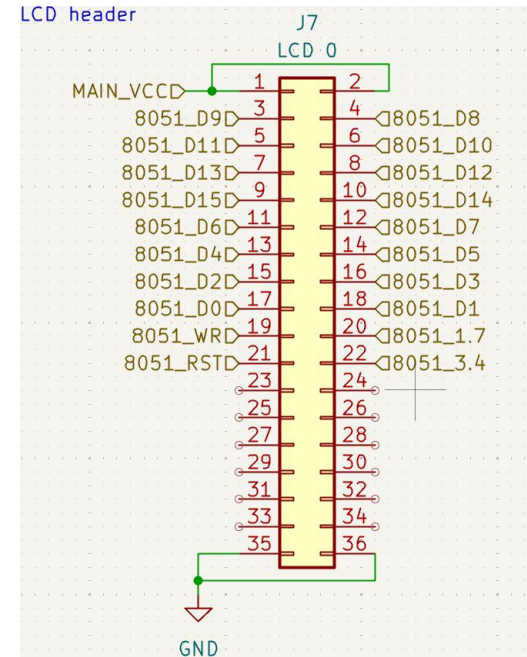
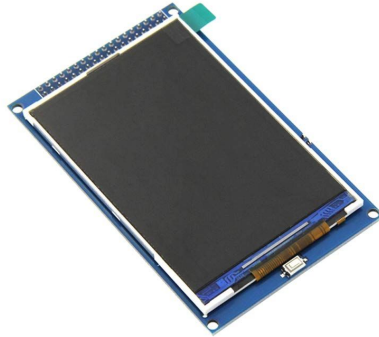
x2

LCD interfacing

No proper reference manual

Chip manual did not have init

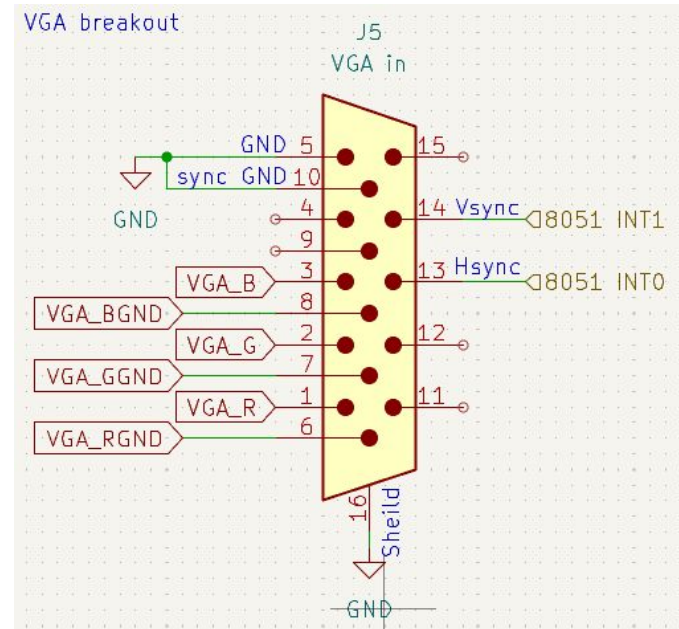
8 bit mode?



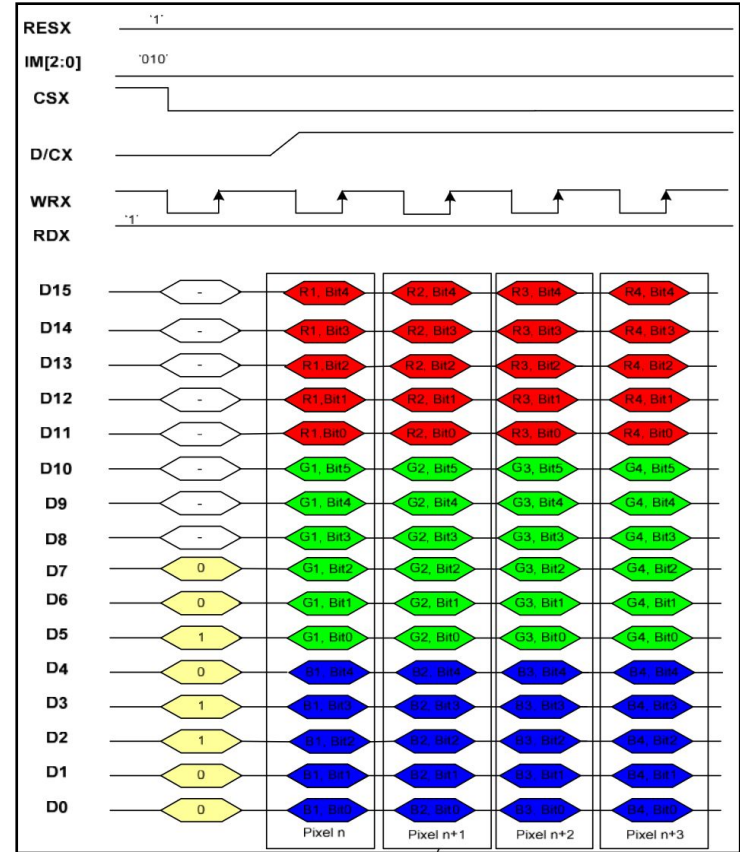
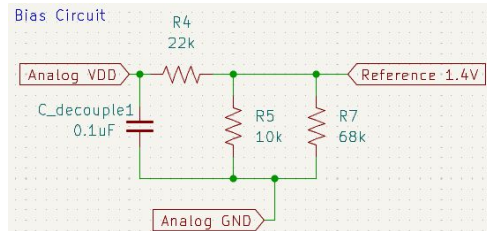
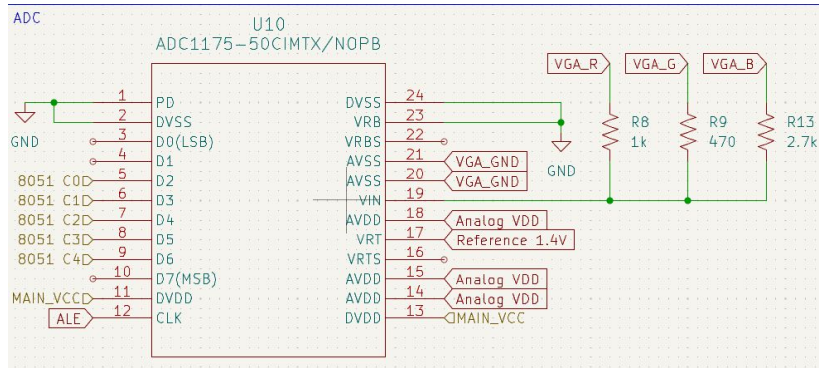
VGA Interfacing

Analog Decoupling

Return lines for RGB



ADC interfacing

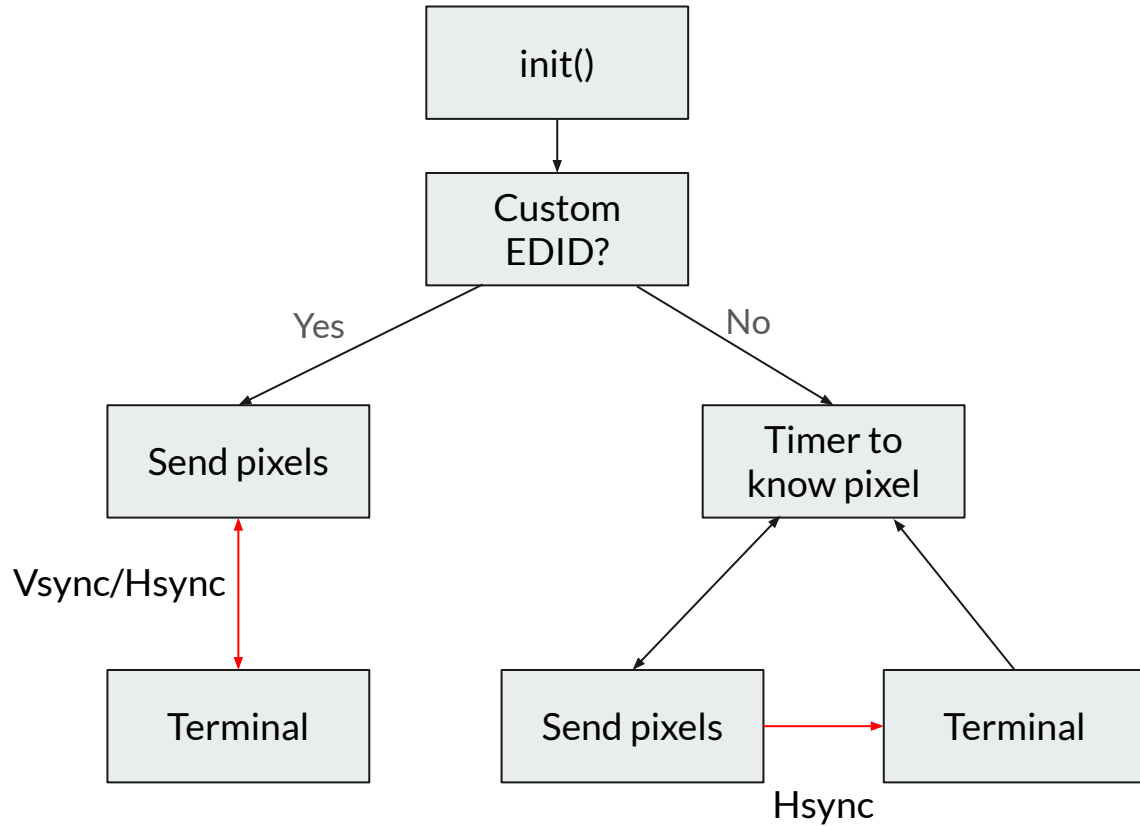


Software

Minimize clock cycles per write

Write the correct pixel

Custom EDID

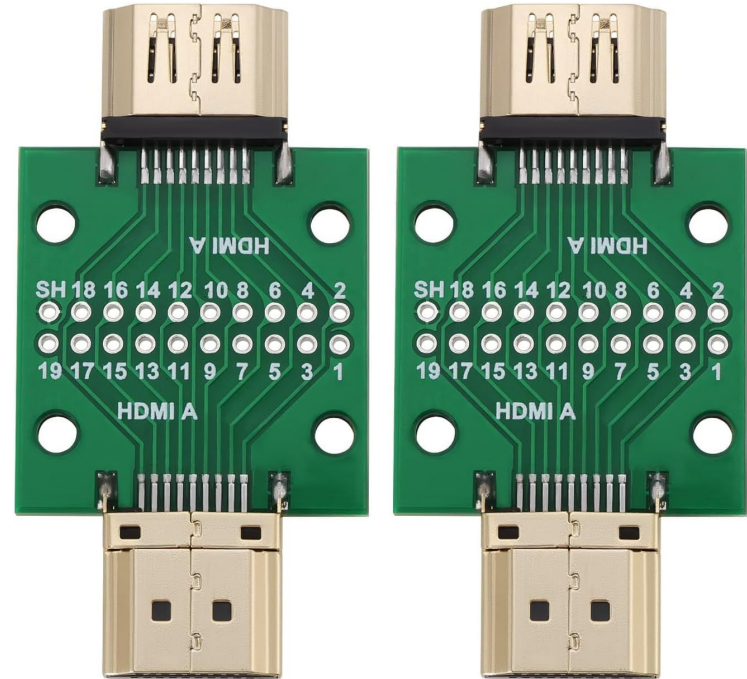


EDID stuff

I2C module

256 bytes

HDMI → VGA has EDID chip





Issues

Writing

LCD init

ADC reference

ADC clock

Buying stuff.



Where I am now

1x LCD working

Reading values from resistor network into ADC

ADC has no clock yet

Can see EDID, can verify it

Need to test timing to get custom EDID timing